

10.1 Practical biology and investigative skills

The requirements at A2 level require further progression by the student as evidenced by the submission of an individual investigation in Unit 6.

Teachers should make the practical and investigative skills criteria available to students to enable them to understand what is expected of them.

Students will further develop their practical skills, whichever approach (context or concept) has been taken. Students will carry out the recommended core practicals in Units 4 and 5 and their individual investigations.

Students will carry out practical work during the GCE Biology A2 course, which will be verified by the teacher using the criteria below and submitted to Edexcel using a verification of practical skills record.

The teacher will verify student's ability to:

- a Use apparatus skilfully and safely
 - i Apparatus and materials are handled correctly and safely and manipulative techniques are used in an appropriate and safe manner.
 - ii The practical work is carried out in an organised, methodical and safe manner, with due consideration of the wellbeing of living organisms and the environment.
- b Produce and record reliable and valid results
 - i Measurements and observations are made with precision and recorded in a structured manner; variables are identified and the validity and reliability of results are justified.
 - ii Possible systematic errors and random errors in generating results are identified and explained.
- c Present and analyse data
 - i Use appropriate methods to analyse results, present data and identify trends, patterns and/or observations.
 - ii Any apparent anomalies and inconsistencies are described, the methodology is evaluated and suggestions are made to improve or further the work of the investigation.

Other practical-related skills, including analysis and evaluation of data may be assessed in the externally assessed components.

Edexcel reserves the right to:

- make it a requirement that centre staff undertake appropriate training to ensure the correct application of the verification process of practical skills
- arrange a centre visit to inspect procedures for the verification of practical skills.

Students are expected to follow the conventions for the collection and presentation of data as set out in the following document issued by the Institute of Biology:

Biological Nomenclature — Standard Terms and Expression Used in the Teaching of Biology, 3rd Edition (Institute of Biology, 2000) ISBN 0900490365.

Edexcel will issue centres with exemplar practical assessment material for Unit 6, Individual Investigation.